

COURSE STRUCTURE

Course Code	CSE0PM07A				
Course Category	Program Major				
Course Title	Competitive Programming				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	0	-	4	-	4
Credits	0	-	2	-	2
Assessment Schema Code	PJ				
Prerequisites: Algorithms and Data structures , C++					
Course Objectives: - To gain the knowledge of Various types of Data Structure - To introduce various techniques for representation of the data in the real world. - To choose appropriate data structure as applied to specified problem definition. - To design various applications based on different data structures. - Use of programming language for application development.					
Course Outcomes: After completion of this course students will be able to: - Understand various concepts of algorithms analysis, recursion - Implement various categories of Arrays and Hashing. - Compare and analyse different searching and sorting algorithms and string concepts. - Implement the Linear data structures and develop applications using it. - Implement the Non-Linear data structures and develop applications using it.					
Course Contents: Unit 1: Introduction					

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for Data Structures, Bitwise Operators for Bit Manipulation, Introduction to Recursion Magic, Tower of Hanoi Problem, Josephus Problem, Subset Sum Problem using Recursion, Printing Permutations

Unit 2: Concept of Array & Hashing

Introduction, Types of Array, Vector in C++, Operations on Arrays, Finding largest Element in an Array, Finding Second Largest Element in Array, Check if an Array is Sorted or not, Reversing an Array, Removing duplicates from a sorted array, Moving Zeros to End, Rotating an Array to left by One place, Leaders problem in an Array, Maximum Difference Problem with Order, Count number of occurrences in a Sorted Array, Trapping Rain Water problem, Finding Maximum subarray sum, Longest Even Odd Subarray, Finding Majority Element in an Array, 2D vector in C++, Arranging Matrix elements in snake Pattern, Boundary elements Traversals of Matrix, Print Transpose of a Matrix, Matrix Rotation

Hashing Concepts - Applications of Hashing, Mapping Records to Corresponding Key values- Direct Address Table, Hashing Functions, Collision Resolution techniques, Introduction and Implementation Separate Chaining, Handling Collision in Hashing (Open Addressing), Handling Collision in Hashing(Double Hashing), Implementation of Open Addressing Strategy, Open Addressing vs Chaining

Unit 3: Searching, Sorting & String

Iterative Binary Search, Recursive Binary Search, Finding Index of first Occurrence of an element in Sorted array, Finding Index of last Occurrence of an element in Sorted array, Count Occurrences of given element in Sorted array, Count number of 1s in a Binary Array, Search element in sorted & Rotated Array, Peak Element, Working on Two Pointer Algorithm, To Find Median of two sorted arrays, Allocate Minimum Pages using Binary Search, Sort in C++ STL, Introduction to Sorting Algorithm, Intersection of two sorted arrays, Find Kth smallest element, Chocolate Distribution Problem, Minimum Difference in an Array, Meeting the maximum guests

Basics of String - Strings using C++, String Palindrome, Check for Subsequence of String, Check for Anagram of String, Check for Leftmost Replication of Character, Check for Leftmost Non-replicating Element, Rearrangement of words from last word to first word in a given string, Verification of String Rotations, Finding of Longest Substring with Distinct Characters, Finding Lexicographic Rank of a String maximum guests

Unit 4: Linear Data Structures

Introduction to Stack, Implementation of Stack using array in C++, Implementation of Stack using Linked List in C++, Applications of Stack, Concept of Stack in C++ STL, Checking Parenthesis, Find the Closest Greater Element, Find the Next Greater Element, Histogram-Largest Rectangular Area, To find Largest Rectangle with all 1's in a matrix. Introduction to Polish notations, Conversion of Infix to Postfix Expression(Efficient Solution)

Introduction to Queue- Queue Application, Implementation using Array, Implementation using Linked

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List, Concept of Queue in C++ STL, Stack Implementation using queue, Introduction to Deque, Implementation of Deque using Array, Concept of Deque in C++ STL

Unit 5: Non-Linear Data Structures

Introduction to Tree Data Structure, Applications of Tree, Binary Tree, Tree Traversal, Print Nodes at K distance, Breadth First Traversal, Size of Binary Tree, Maximum in Binary Tree, Check for Balanced Binary Tree, Maximum Width of Binary Tree, Count nodes in a Complete Binary Tree, Binary Search Tree(Introduction)- Search in BST using C++ , Insert in BST using C++, Deletion in BST using C++, Set in C++ STL, Map in C++ STL, Find Kth Smallest in BST

Introduction to Graph Data Structure, Applications of Graph, Adjacency Matrix, Adjacency List, Pairs in C++, Vectors of Pair in C++, Construction of undirected graph, Construction of directed graph, Construction of weighted graph, Breadth First Search using Queue, Depth First Search Using Queue Kruskal's Minimum Spanning Tree, Minimum Spanning Tree using Prim's Algorithm, Dijkstra Shortest Path Algorithm, Bellman Ford Shortest Path Algorithm, Floyd-Warshall Algorithm, Sales person Problem

Laboratory Exercises / Practical: Detailed list of experiment. 1 to 12 (Every experiment consist of more than 3 subprograms)

1. Write a C ++ program using STL:

- i. To count the number of words in a string
- ii. To find a matrix's transposition
- iii. To find the Fibonacci Series.
- iv. To find exponential number (M to power N)
- v. Consider below given code and find out the time complexity of it

```
int try fun (int n)
{
    int cnt = 0;
    for (int k = 0; k < n; k++)
        for (int p = k; p > 0; p--)
            cnt = cnt + 1;
    return cnt;
}
```

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2. Practical Assessment Mathematics and Bit manipulation:

- i. Checking a bit at nth position (set or clear): Given a number N. bit position: n. We have to check whether bit n in a number set or not. If in binary representation of a number a bit at particular position is 1 then it is called as set. Position of set bit should be counted with 0 from LSB side.
- ii. Find Maximum AND value produced by pairwise(any) element from the array: Given an

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- array `ARRAY []` of N elements. We have to find Maximum AND value produced by pairwise(any)element from the array. Operator used is AND which is bitwise & operator.
- iii. Check whether number is even or odd: Given a number N . We have to check whether a number is even or odd using bitwise & operator.
 - iv. Given a positive integer P . Your task is to find number of Digits in Factorial
 - v. Given an integral number P . The task is to find the count of digits present in this number.

3. Practical Assessment Recursion Problem Solving Skill:

- i. You are given a number N , the task is to print all the numbers from N to 1 using recursion.
- ii. You are given a number N , the task is to find the sum of its individual digits using recursion
- iii. You are given a number N and it's power value as m then the task is to find power m of the number N using recursion
- iv. You are given a string, print all permutations of it. Permutation is the process of rearranging the items in an ordered list S so that each item corresponds to another element in S . The string of length p is $p!$ Permutation.
- v. Given a set as an array and a sum, we need to find out count of subsets of this array whose sum is equal to given sum.

4. Practical Assessment Array:

- i. Finding Second Largest Element in Array
- ii. Check if an Array is Sorted or not
- iii. Move all zeroes to end of array
- iv. Rotating an Array to left by One place
- v. Leaders problem in an Array: Accept integer array an element is called as leader if it is greater than all the elements to its right side and the rightmost element is always a leader. Element is leader when all elements of its right are smaller even equals are not allowed for element to be a leader.
- vi. Print Transpose of a Matrix
- vii. Arranging Matrix elements in snake Pattern: Given a matrix of size $n \times n$. print the matrix elements in the snake pattern.

5. Practical Assessment Introduction to Hashing

- i. Use the Separate chaining strategy to fill up the hash table in this question. Fill the items of the array into a hash table of the specified size provided an array (containing of unique integers) and the hashtable size.
- ii. Linear Probing in Hashing, Given a hash table size and an array of integers. To handle collisions, put the elements of the array into a hash table using linear probing. Colliding items must be mapped to the $[(value+1) \% hashSize]$ position while duplicate elements must be mapped to the same position in the hash table.
- iii. You must insert the members of an array (containing of distinct integers) into a hash table of the specified size, given the array and the size of the hash table.

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- iv. Given a hash table size and an array of integers. Identical items must be mapped to the same place in the hash table, whereas colliding elements must be mapped to the opposite position. $[(value+1) \% hashSize1]$ position.

6. Practical Assessment Introduction to Searching

- Recursive Binary Search: Search an element by tripling it after every successful occurrence
- Finding index of first Occurrence of an element in a Sorted array
- Peak Element: Find index of any peak element in given array with time complexity $O(\log n)$
- Working on Two Pointer Algorithm: Given an array which is sorted with N elements then find pair of elements whose sum is equal to X.

7. Practical Assessment Introduction to Sorting

- Implement sort () function using C++ STL.
- Bubble Sort using STL in C++
- Sort elements using Insertion Sort
- Implement Sorting Operations-Intersection of two sorted arrays
- Kth Smallest Element
- Find triplets with zero sum: Given is an array of n numbers, m_arr []. Check to see if it includes triplets with a sum of zero.

8. Practical Assessment String Operations:

- You are given a string S1, the task is to check if the string S1 is Palindrome or not.
- Two strings are provided to you. as srcstring and findstring, the task is to find if the findstring is present in the srcstring. Return -1 if findstring is not found.
- You are given two strings as str1 and str2, then the task is to find out whether string str1 is an anagram of string str2.
- You are given a string S1, the task is to print leftmost repeating character in the string, return -1 if not found.
- You are given a string S1. The task is to reverse the order of words in the given string i.e. the first word should be last and last word as first word. Note that do not reverse character of every word.

9. Practical Assessment Stack

- Apply stack using an array, to implement a stack, you must utilise the class complete the push () and pop() operations.
- Implement Linked List with Stack, you are provided a linked list, and you must use this linked list to implement the stack's push and pop functionalities.
- Paragraph Checker, suppose x is an expression string. Check to see if the pairs and ordering of

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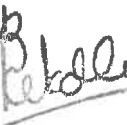


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"", "(", ")", "[", "]" are accurate in the expression.

- iv. Postfix Expression Evaluation, given string S representing a postfix expression, the task is to evaluate the expression and find the final value. Operators will only include the basic arithmetic operators like *, /, + and -.
- v. Infix to Postfix, Given a string expression with the infix form str. This infix expression should be changed to a postfix expression.

10. Practical Assessment Queue

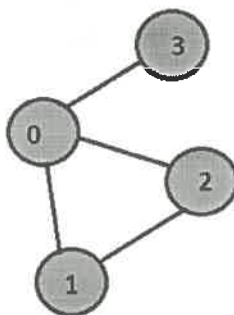
- i. Implement a Queue using an Array.
- ii. Prepare a CPP code to illustrate Queue1 in Standard Template Library (STL)
- iii. Given a number of integers, Enter them to the queue and find the size of the queue1 without using the size function.
- iv. Having the ability to enqueue and dequeue actions from both ends, Implement a double-ended queue.

11. Practical Assessment Binary Tree

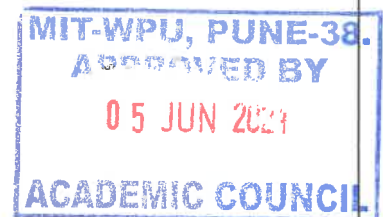
- i. Create and return the binary tree, given two integer arrays, preorder and inorder, where preorder denotes a binary tree's preorder traversal and inorder denotes the tree's inorder traversal.
- ii. Write a menu driven 'C' program to implement Tree Traversal on Binary Tree.
- iii. Create and return the binary tree for given data, find its inorder traversal.
Input: [1,2,3,4,5,6]
- iv.

12. Practical Assessment Graph

- i. For the given below graph find the adjacency matrix
Example



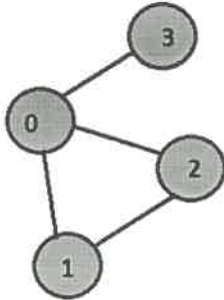
- ii. For the given below graph find the adjacency list
Example



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- iii. Write a program for undirected graph to count the total no of edges

Learning Resources:

Text Books/ Reference Books:

- Reema Thareja, "Data Structures using C" - Oxford University, Second Edition, ISBN -0-19-809930-4
- Treamblay, Sorenson, "An introduction to data structures with applications" Tata McGraw Hill, Second Edition, ISBN: 9780070651579
- Horowitz, S. Sahani, S. Anderson-Freed, "Fundamentals of Data Structures in C", Universities Press, ISBN ISBN 0929306406, Publication year 2008
- Yedidyah Langsam, Moshe J Augenstein, Aron M Tenenbaum, "Data Structures using C and C++", Pearson Education, ISBN 81-317-0328-2

Supplementary Reading:

Web Resources:

Weblinks:

- https://www.tutorialspoint.com/data_structures_algorithms
- <https://www.studytonight.com/data-structures>
- <https://www.khanacademy.org/computing/computer-science/algorithms>
- https://www.geeksforgeeks.org/data_structures

MOOCs: Online courses for self-learning

- iv. <https://www.udemy.com/learning-data-structures-and-algorithms/>
- v. <https://nptel.ac.in/courses/106103069/>

Pedagogy:

- vi. Group activity
- vii. Power point presentations
- viii. Videos
- ix. Demonstrations
- x. Systematic use of group work and project-based learning.



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